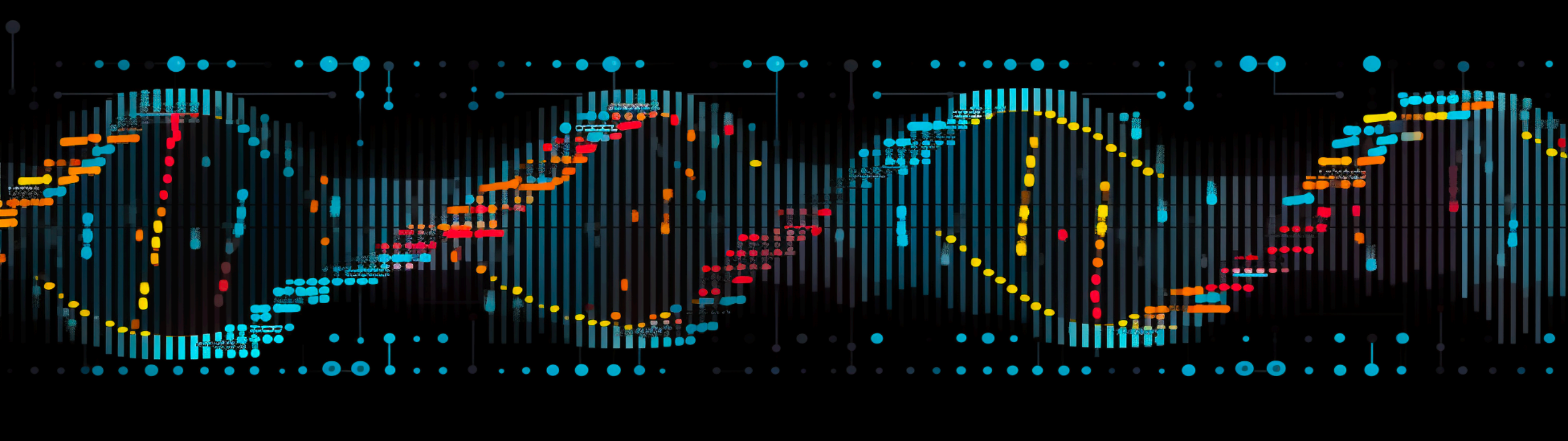


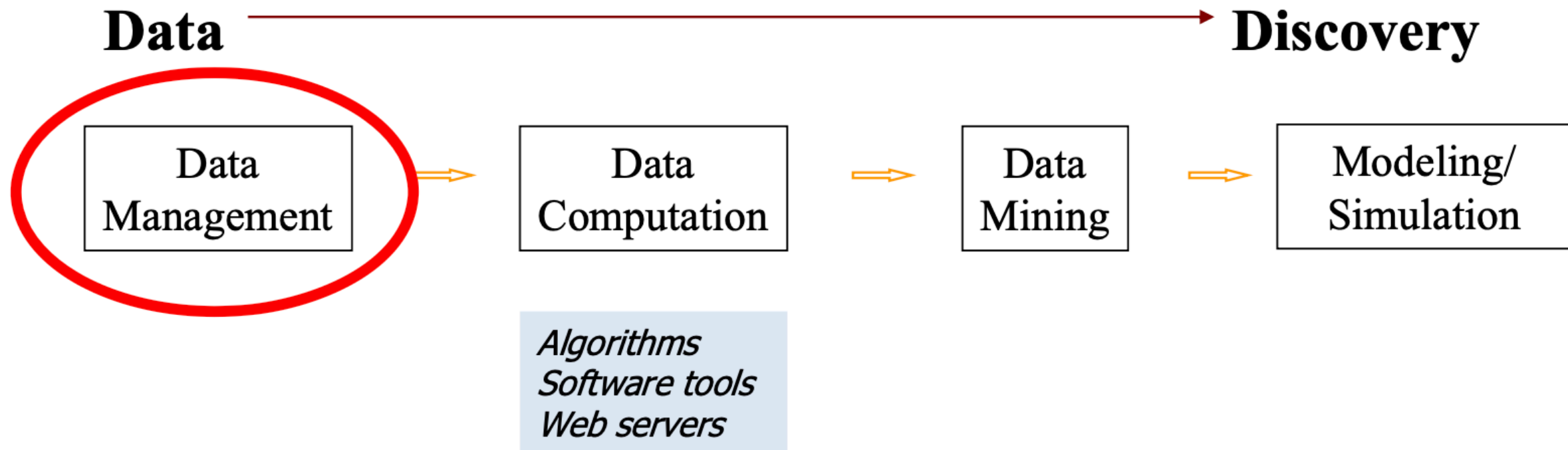


Introduction to bioinformatics

Chapter3 Data and Resource Part1- Gene Ontology and KEGG

Jiaxing Chen

The –informatics in Bioinformatics



How can a computer know?!

wnt1

wnt-1

int1

WINGLESS-TYPE MMTV INTEGRATION SITE FAMILY, MEMBER 1

WINGLESS-TYPE MMTV INTEGRATION SITE FAMILY

wingless

wg

We need to define it for the computer!

- Entity: name, synonyms, acronyms, properties
- Relationship: e.g., is-a

hierarchical, common, controlled vocabulary

the use of predefined, authorized terms that have been preselected by the designer of the vocabulary, in contrast to free natural language vocabularies.

Ontology

A specification of a conceptualization

<http://www-ksl.stanford.edu/kst/what-is-an-ontology.html>

A set of concepts within a domain, defined by a shared vocabulary to denote the types and properties of the concepts as well as the relationships between the concepts

In philosophy: the study of the nature of being, becoming, existence, or reality, as well as the basic categories of being and their relations

What does an ontology enable?

Communication: to be able to communicate unambiguously

e.g., to understand across different groups' annotations of various genomes

Computation: to be able to represent knowledge in a computable form

e.g., to represent literature and data in a structured form to enable automated analyses by computer programs

Discovery of Patterns: to be able to traverse different hierarchies

e.g., to go above a set of individual genes to find the larger functional categories or pathways involved, with a bird's-eye view.

Open Biomedical Ontologies (OBO)

Gene Ontology

Anatomical Entity Ontology

Disease Ontology

Sequence Ontology

System Biology Ontology

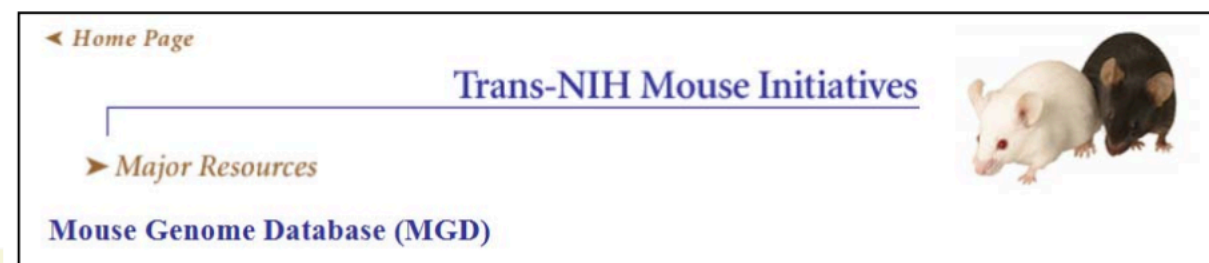
.....

<http://www.obofoundry.org/>

Gene Ontology (GO)

Ashburner *et al.*, *Nat. Genet.*, 2000

- The project began as a collaboration between three model organism genome databases: FlyBase, *Saccharomyces* Genome Database (SGD), and Mouse Genome Database (MGD) in 1998.
- Defines a structured, common, controlled vocabulary to describe attributes of genes and gene products across organisms.



http://www.nih.gov/science/models/mouse/resources/mgd.html

Three categories

Molecular Function = elemental activity/task

the tasks performed by individual gene products; examples are carbohydrate binding and ATPase activity

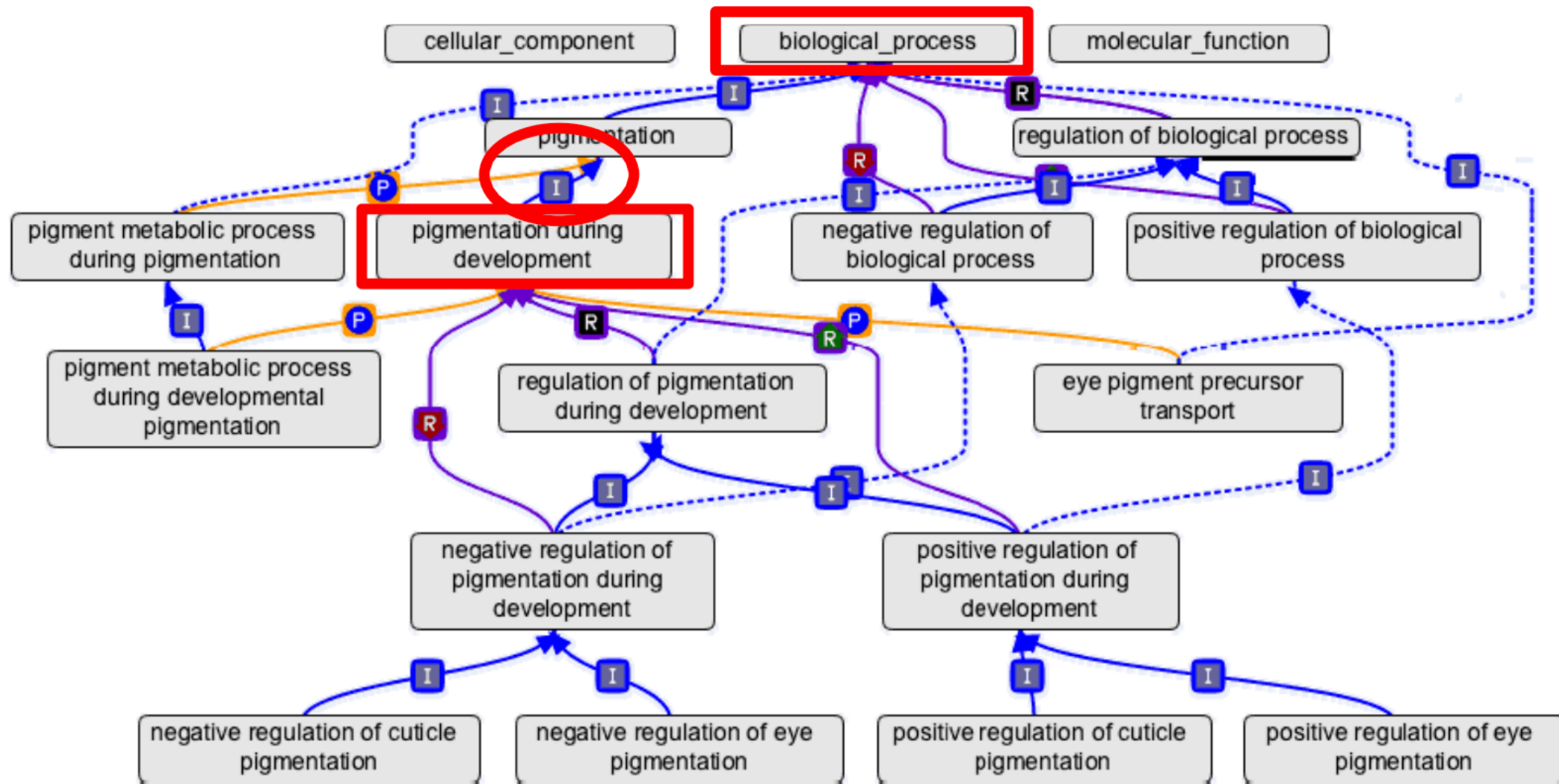
Biological Process = biological goal or objective

broad biological goals, such as mitosis or purine metabolism, that are accomplished by ordered assemblies of molecular functions

Cellular Component = location or complex

Subcellular structures, locations, and macromolecular complexes; examples include nucleus, telomere, and RNA polymerase II holoenzyme

GO structure: **Directed Acyclic** Graph (DAG)



<http://www.geneontology.org/images/diag-ontology-graph.gif>

How to store this graph in a computer?

OBO File Format

[Term]

id

name

namespace

def

synonym

is_a

OBO format example

```
[Term]
id: GO:0000001
name: mitochondrion inheritance
namespace: biological_process
def: "The distribution of mitochondria, including the mitochondrial genome, into daughter cells after mitosis mediated by interactions between mitochondria and the cytoskeleton." [GOC:mcc, PMID:10873824, PMID:11389764]
synonym: "mitochondrial inheritance" EXACT []
is_a: GO:0048308 ! organelle inheritance
is_a: GO:0048311 ! mitochondrion distribution

[Term]
id: GO:0000002
name: mitochondrial genome maintenance
namespace: biological_process
def: "The maintenance of the structure and integrity of the mitochondrial genome; includes replication and segregation of the mitochondrial chromosome." [GOC:ai, GOC:vw]
is_a: GO:0007005 ! mitochondrion organization

[Term]
id: GO:0000003
name: reproduction
namespace: biological_process
```

XML format

go:term

go:accession

go:name

go:synonym

go:definition

go:isa

go:dbxref

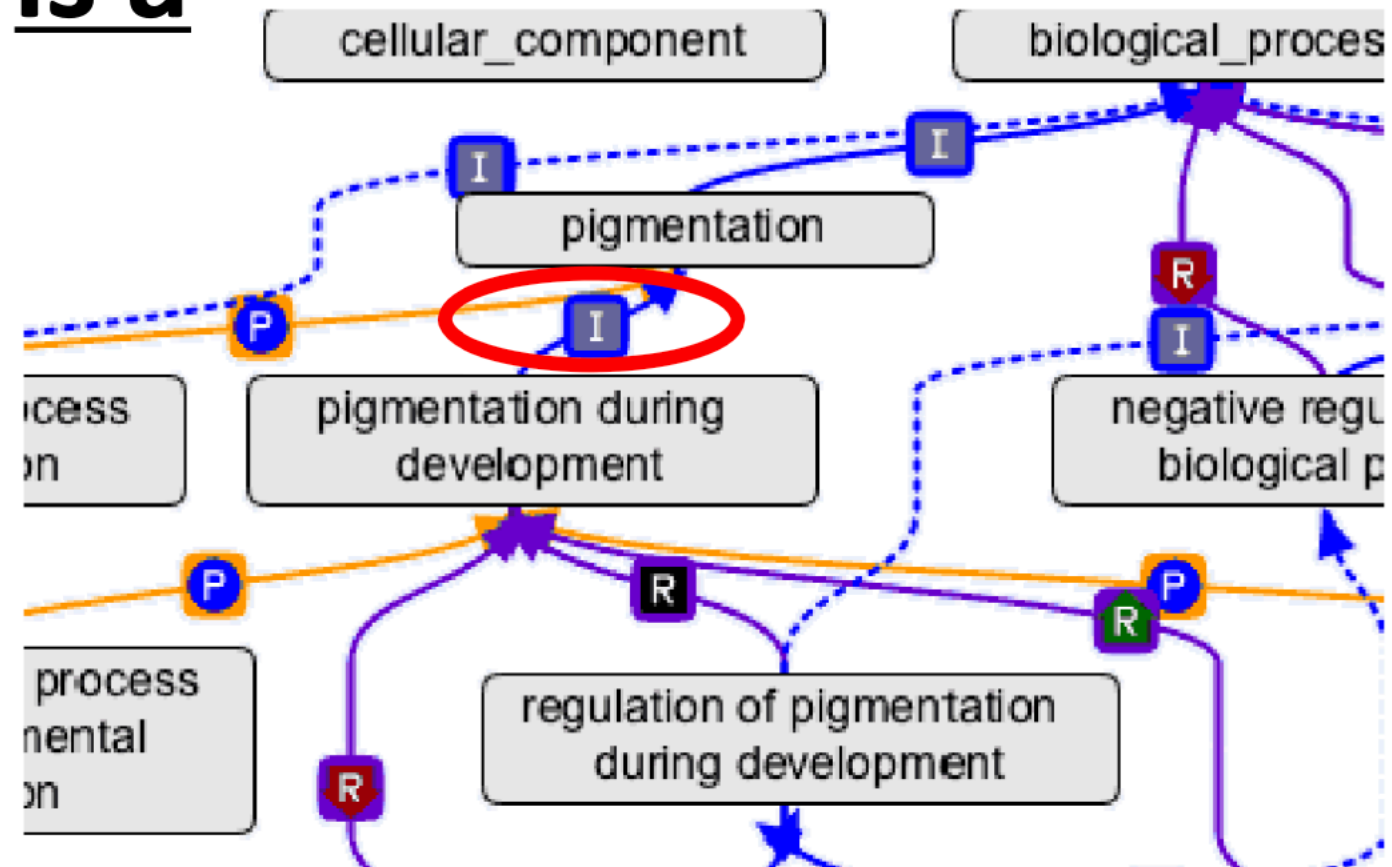
RDF-XML format example

```
<go:term rdf:about="http://www.geneontology.org/go#GO:0000001">
  <go:accession>GO:0000001</go:accession>
  <go:name>mitochondrion inheritance</go:name>
  <go:synonym>mitochondrial inheritance</go:synonym>
  <go:definition>The distribution of mitochondria, including the mitochondrial genome, into daughter cells after mitosis or meiosis, mediated by interactions between mitochondria and the cytoskeleton.</go:definition>
  <go:is_a rdf:resource="http://www.geneontology.org/go#GO:0048308" />
  <go:is_a rdf:resource="http://www.geneontology.org/go#GO:0048311" />
</go:term>

<go:term rdf:about="http://www.geneontology.org/go#GO:0000002">
  <go:accession>GO:0000002</go:accession>
  <go:name>mitochondrial genome maintenance</go:name>
  <go:definition>The maintenance of the structure and integrity of the mitochondrial genome; includes replication and segregation of the mitochondrial chromosome.</go:definition>
  <go:is_a rdf:resource="http://www.geneontology.org/go#GO:0007005" />
  <go:dbxref rdf:parseType="Resource">
    <go:database_symbol>InterPro</go:database_symbol>
    <go:reference>IPR009446</go:reference>
  </go:dbxref>
  <go:dbxref rdf:parseType="Resource">
    <go:database_symbol>InterPro</go:database_symbol>
    <go:reference>IPR016610</go:reference>
  </go:dbxref>
```

GO relationship: is a

“B is a A”: B is a subtype of A



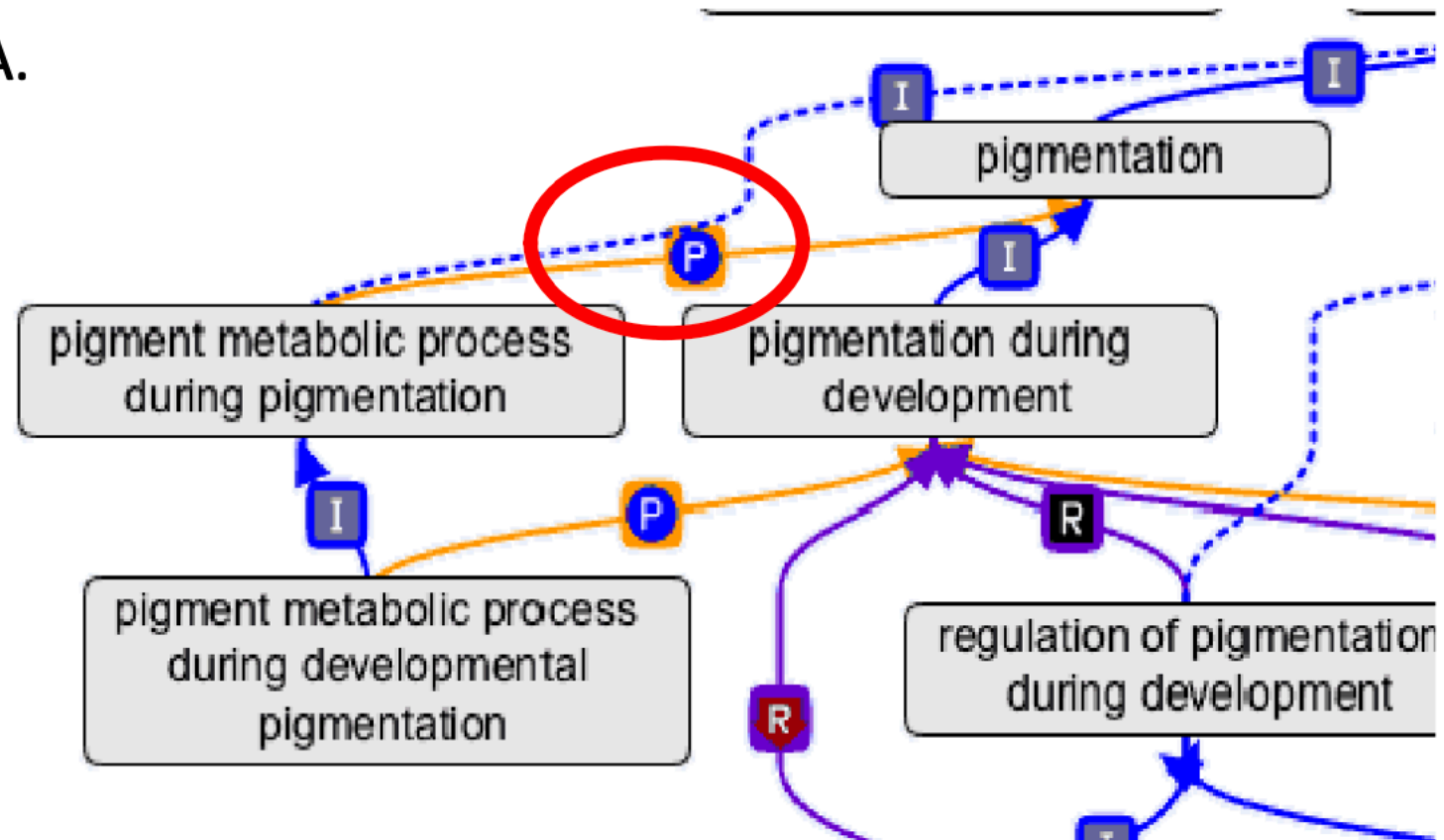
Examples:

mitochondrion inheritance is an organelle inheritance

pigmentation during development is a pigmentation

GO relationship: part of

B **part of** A: B is a part of A.



Examples:

ribosomal large subunit assembly is **part of** ribosome assembly

pigment metabolic process during pigmentation is **part of** pigmentation

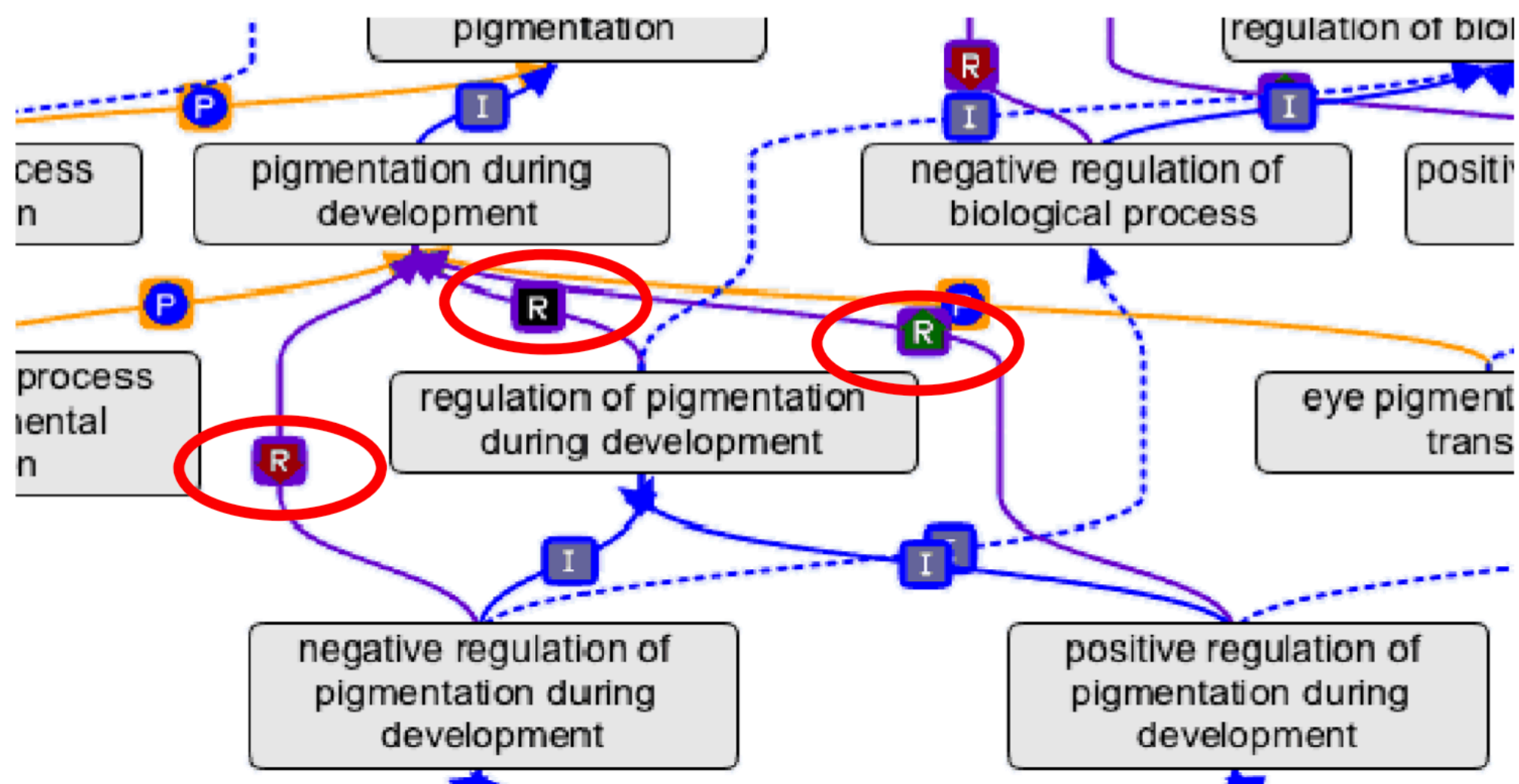
GO relationship: regulates

B regulates A

Sub-relationships:

positively regulates

negatively regulates



Reasoning over the relationships

A is a B ◦ B is a C → A is a C

A is a B ◦ B part of C → A part of C

A part of B ◦ B is a C → A part of C

A part of B ◦ B part of C → A part of C

A is a B ◦ B regulates C → A regulates C

A regulates B ◦ B is a C → A regulates C

A regulates B ◦ B part of C → A regulates C

Reasoning over the relationships

A is a B ◦ B positively regulates C → A positively regulates C

A positively regulates B ◦ B is a C → A positively regulates C

A positively regulates B ◦ B part of C → A positively regulates C

A is a B ◦ B negatively regulates C → A negatively regulates C
A negatively regulates B ◦ B is a C → A negatively regulates C
A negatively regulates B ◦ B part of C → A negatively regulates C

Current release 2024-01-17: 42,442 GO terms | 7,655,937 annotations
1,537,348 gene products | 5,387 species ([see statistics](#))

THE GENE ONTOLOGY RESOURCE

The mission of the GO Consortium is to develop a comprehensive, **computational model of biological systems**, ranging from the molecular to the organism level, across the multiplicity of species in the tree of life.

The Gene Ontology (GO) knowledgebase is the world's largest source of information on the functions of genes. This knowledge is both human-readable and machine-readable, and is a foundation for computational analysis of large-scale molecular biology and genetics experiments in biomedical research.

GO Enrichment Analysis ?

Powered by PANTHER

Your gene IDs here...

biological process

Homo sapiens

Examples

Launch >

Hint: can use UniProt ID/AC, Gene Name, Gene Symbols, MOD IDs

Search GO term or Gene Product in AmiGO ...

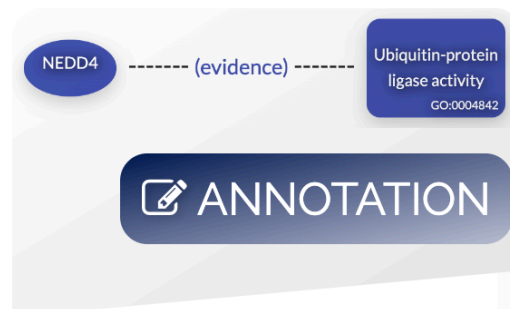
☒ Any ☐ Ontology ☐ Gene Product

<https://www.geneontology.org/>



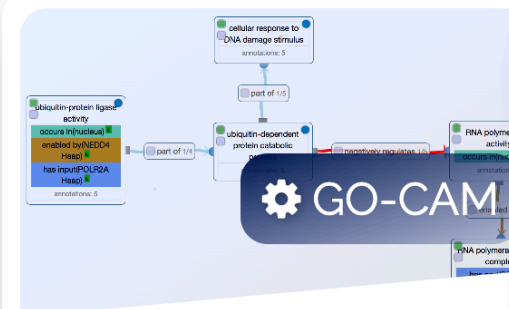
The network of biological classes describing the current best representation of the “universe” of biology: the molecular functions, cellular locations, and processes gene products may carry out.

[GO Ontology Overview](#)
[Browse in AmiGO](#)
[Download](#)



Statements, based on specific, traceable scientific evidence, asserting that a specific gene product is a real exemplar of a particular GO class.

[GO Annotations Overview](#)
[Browse in AmiGO](#)
[Download](#)



GO Causal Activity Model (GO-CAM) provides a structured framework to link standard GO annotations into a more complete model of a biological system.

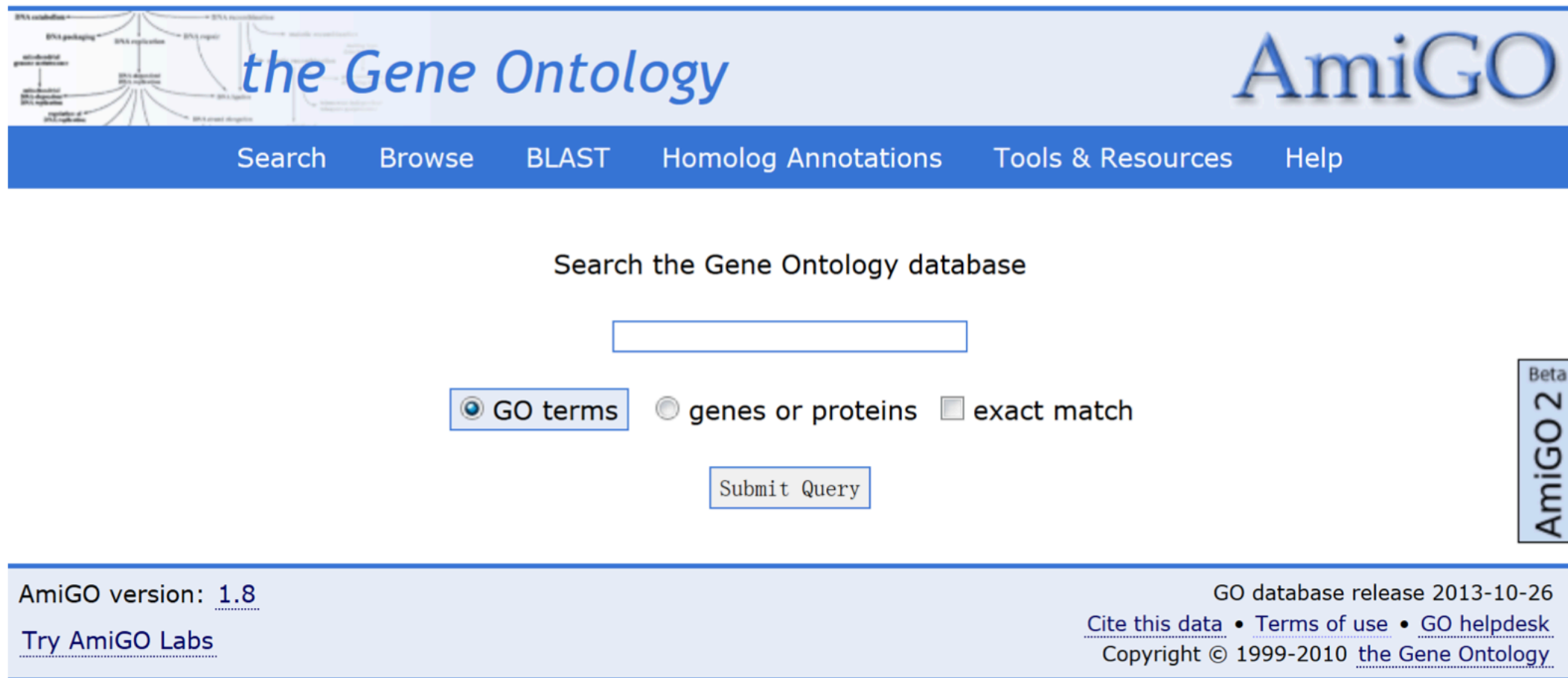
[GO-CAM Overview](#)
[Browse GO-CAMs](#)
[Download](#)



Tools to curate, browse, search, visualize and download both the ontology and annotations. Includes bioinformatic guides (Notebooks) and simple API access to integrate the GO into your research.

[GO Tools Overview](#)
[GO APIs Guide](#)
[GO GitHub](#)

AmiGO: GO browser and search engine



The image shows the AmiGO web interface. At the top, there is a header with the Gene Ontology logo on the left and the AmiGO logo on the right. Below the logos is a navigation bar with links: Search, Browse, BLAST, Homolog Annotations, Tools & Resources, and Help. The main content area has the text "Search the Gene Ontology database" above a search input field. Below the input field are three radio buttons: "GO terms" (selected), "genes or proteins", and "exact match". A "Submit Query" button is located below the radio buttons. On the right side of the main content area, there is a vertical box labeled "Beta" and "AmiGO 2". At the bottom, there is a footer with the text "AmiGO version: 1.8" and "Try AmiGO Labs" on the left, and "GO database release 2013-10-26", "Cite this data", "Terms of use", "GO helpdesk", and "Copyright © 1999-2010 the Gene Ontology" on the right.

the Gene Ontology

AmiGO

Search Browse BLAST Homolog Annotations Tools & Resources Help

Search the Gene Ontology database

☒ GO terms ☐ genes or proteins ☐ exact match

Submit Query

Beta
AmiGO 2

AmiGO version: 1.8
Try AmiGO Labs

GO database release 2013-10-26
Cite this data • Terms of use • GO helpdesk
Copyright © 1999-2010 the Gene Ontology

Browse GO

the Gene Ontology

AmiGO

Search Browse BLAST Homolog Annotations Tools & Resources Help

Search GO ☐ terms ☐ genes or proteins ☐ exact match

Tree Browser

▼ Filter tree view ?

Filter by ontology

Ontology

All
biological process
cellular component
molecular function

Filter Gene Product Counts

Data source

All
ASAP
AspGD
CGD

Species

All
Arabidopsis thaliana
Aspergillus fumig...
Aspergillus fumig...

View Options

Tree view

☒ Full ☐ Compact

all : all [699004 gene products] 

☐  GO:0008150 : biological_process [543249 gene products] 

☐  GO:0022610 : biological adhesion [8711 gene products] 

☐  GO:0051825 : adhesion to other organism involved in symbiotic interaction [970 gene products] 

☐  GO:0044406 : adhesion to host [954 gene products] 

☐  **GO:0075001 : adhesion of symbiont infection structure to host [1 gene product] **

☐  GO:0075003 : adhesion of symbiont appressorium to host [0 gene products]

☐  GO:0075002 : adhesion of symbiont germination tube to host [0 gene products]

☐  GO:0075196 : adhesion of symbiont haustorium mother cell to host [0 gene products]

☐  GO:0075070 : adhesion of symbiont hyphopodium to host [0 gene products]

☐  GO:0075069 : adhesion of symbiont infection cushion to host [0 gene products]

☐  GO:0075004 : adhesion of symbiont spore to host [0 gene products]

Actions...

Last action: Opened

GO:0075001

Graphical View

Permalink

Download...

OBO

RDF/XML

GraphViz dot

Search for GO terms or genes

the Gene Ontology **AmiGO**

Search Browse BLAST Homolog Annotations Tools & Resources Help

Search GO ☐ GO terms ☒ genes or proteins ☐ exact match

Gene Products **the Gene Ontology** **AmiGO**

Search GO ☐ terms ☒ genes or proteins ☐ exact match

11 results for MeCP2 in genes or proteins fields symbol, full name(s) and synonym(s)

Filter search results

Filter Gene Products by

Gene Product Type: All, complex, gene, gene product

Data source: All, ASAP, AspGD, CGD

Species: All, Arabidopsis thaliana, Aspergillus fumigatus, Aspergillus fumigatus...

Filter Gene Products by Ontology: All, biological process, cellular component, molecular function

Results are sorted by **relevance**. To change the sort order, click on the column headers. ★ indicates that the gene product is a member of a **homolog set**. Click on the gene product to view its details.

Select all Clear all Perform an action with this page's selected gene products... Go!

rel ↓	Symbol, full name	78 associations
☐	Mecp2 ★ methyl CpG binding protein 2	BLAST
☐	Mecp2 ★ methyl CpG binding protein 2	BLAST
☐	mecp2 ★ methyl CpG binding protein 2	BLAST

mitochondrion inheritance

Term Information

Accession: GO:0000001

Ontology: Biological Process

Synonyms: exact: mitochondrial inheritance

Definition: The distribution of mitochondria, including the mitochondrial genome, into daughter cells after mitosis or meiosis, mediated by interactions between mitochondria and the cytoskeleton.

Source: GOC:mcc, PMID:10873824, PMID:11389764

Comment: None

Subset: None

Community: Add usage comments for this term on the GONUTS wiki.

Term Neighborhood for mitochondrion inheritance (GO:0000001)

Filter lineage gene product counts

Data source: All, ASAP, AspGD, CGD

Species: All, Arabidopsis thaliana, Aspergillus fumigatus, Aspergillus fumigatus...

Ancestors and Children

Inferred Tree View Graph View Other Views Downloads Mappings

subject	relation	object	annotations
mitochondrion inheritance	is_a (inferred)	biological_process (GO:0008150)	543249
mitochondrion inheritance	is_a (inferred)	localization (GO:0051179)	73595
mitochondrion inheritance	is_a (inferred)	cellular_component_organization_or_biogenesis (GO:0071848)	65638
mitochondrion inheritance	is_a (inferred)	cellular_localization (GO:0051541)	26549
mitochondrion inheritance	is_a (inferred)	cellular_process (GO:0009987)	302930

Download the ontology

Files are available in the following formats:

- [OBO 1.4](#): human- and machine-readable; can be opened in any text editor
- [OWL](#): can be opened with [Protégé](#).
- Three versions of the ontology are available, as described below.

Name	Description	Permanent URL	Download link
go-basic.obo	The basic version of the GO, filtered such that the graph is guaranteed to be acyclic and annotations can be propagated up the graph. The relations included are <i>is a</i> , <i>part of</i> , <i>regulates</i> , <i>negatively regulates</i> and <i>positively regulates</i> . This version excludes relationships that cross the 3 GO hierarchies. This version should be used with most GO-based annotation tools.	http://purl.obolibrary.org/obo/go/go-basic.obo	Download go-basic.obo file
go.obo & go.owl	Core ontology. This view includes relationships not in the filtered version of GO including <i>has_part</i> and <i>occurs_in</i> . Many of these relationships may not be safe for propagating annotations across, so this version should not be used with legacy GO tools. This version excludes relationships to external ontologies.	http://purl.obolibrary.org/obo/go.obo / http://purl.obolibrary.org/obo/go.owl	Download go.obo file / Download go.owl file

```
format-version: 1.2
data-version: releases/2024-01-17
subsetdef: chebi_ph7_3 "Rhea list of ChEBI terms representing the major species at pH 7.3."
subsetdef: gocheck_do_not_annotate "Term not to be used for direct annotation"
subsetdef: gocheck_do_not_manually_annotate "Term not to be used for direct manual annotation"
subsetdef: goslim_agr "AGR slim"
subsetdef: goslim_aspergillus "Aspergillus GO slim"
subsetdef: goslim_candida "Candida GO slim"
subsetdef: goslim_chembl "ChEMBL protein targets summary"
subsetdef: goslim_drosophila "Drosophila GO slim"
subsetdef: goslim_flybase_ribbon "FlyBase Drosophila GO ribbon slim"
subsetdef: goslim_generic "Generic GO slim"
subsetdef: goslim_metagenomics "Metagenomics GO slim"
subsetdef: goslim_mouse "Mouse GO slim"
subsetdef: goslim_pir "PIR GO slim"
subsetdef: goslim_plant "Plant GO slim"
subsetdef: goslim_pombe "Fission yeast GO slim"
subsetdef: goslim_prokaryote "GO subset for prokaryotes"
subsetdef: goslim_synapse "synapse GO slim"
subsetdef: goslim_yeast "Yeast GO slim"
synonymtypedef: syngo_official_label "label approved by the SynGO project"
synonymtypedef: systematic_synonym "Systematic synonym" EXACT
default-namespace: gene_ontology
ontology: go

[Term]
id: GO:0000001
name: mitochondrion inheritance
namespace: biological_process
def: "The distribution of mitochondria, including the mitochondrial genome, into daughter cells after mitosis or meiosis, mediated by interactions between mitochondria and the cytoskeleton." [GOC:mcc, PMID:10873824, PMID:11389764]
synonym: "mitochondrial inheritance" EXACT []
is_a: GO:0048308 ! organelle inheritance
is_a: GO:0048311 ! mitochondrion distribution
```

You can download the latest Gene Ontology database from link:
<https://current.geneontology.org/ontology/go-basic.obo>

[Term]
id: GO:0000002
name: mitochondrial genome maintenance
namespace: biological_process
def: "The maintenance of the structure and integrity of the mitochondrial genome; includes replication and segregation of the mitochondrial chromosome." [GOC:ai, GOC:vw]
is_a: GO:0007005 ! mitochondrion organization

[Term]
id: GO:0000003
name: reproduction
namespace: biological_process
alt_id: GO:0019952
alt_id: GO:0050876
def: "The production of new individuals that contain some portion of genetic material inherited from one or more parent organisms." [GOC:go_curators, GOC:isa_complete, GOC:jl, ISBN:0198506732]
subset: goslim_agr
subset: goslim_chembl
subset: goslim_flybase_ribbon
subset: goslim_pir
subset: goslim_plant
synonym: "reproductive physiological process" EXACT []
xref: Wikipedia:Reproduction
is_a: GO:0008150 ! biological_process

[Term]
id: GO:0000005
name: obsolete ribosomal chaperone activity
namespace: molecular_function
def: "OBSOLETE. Assists in the correct assembly of ribosomes or ribosomal subunits in vivo, but is not a component of the assembled ribosome when performing its normal biological function." [GOC:jl, PMID:12150913]
comment: This term was made obsolete because it refers to a class of gene products and a biological process rather than a molecular function.
synonym: "ribosomal chaperone activity" EXACT []
is_obsolete: true
consider: GO:0042254
consider: GO:0044183
consider: GO:0051082

What is a biological pathway?

A series of actions among molecules in a cell that lead to a certain product or a change in a cell.

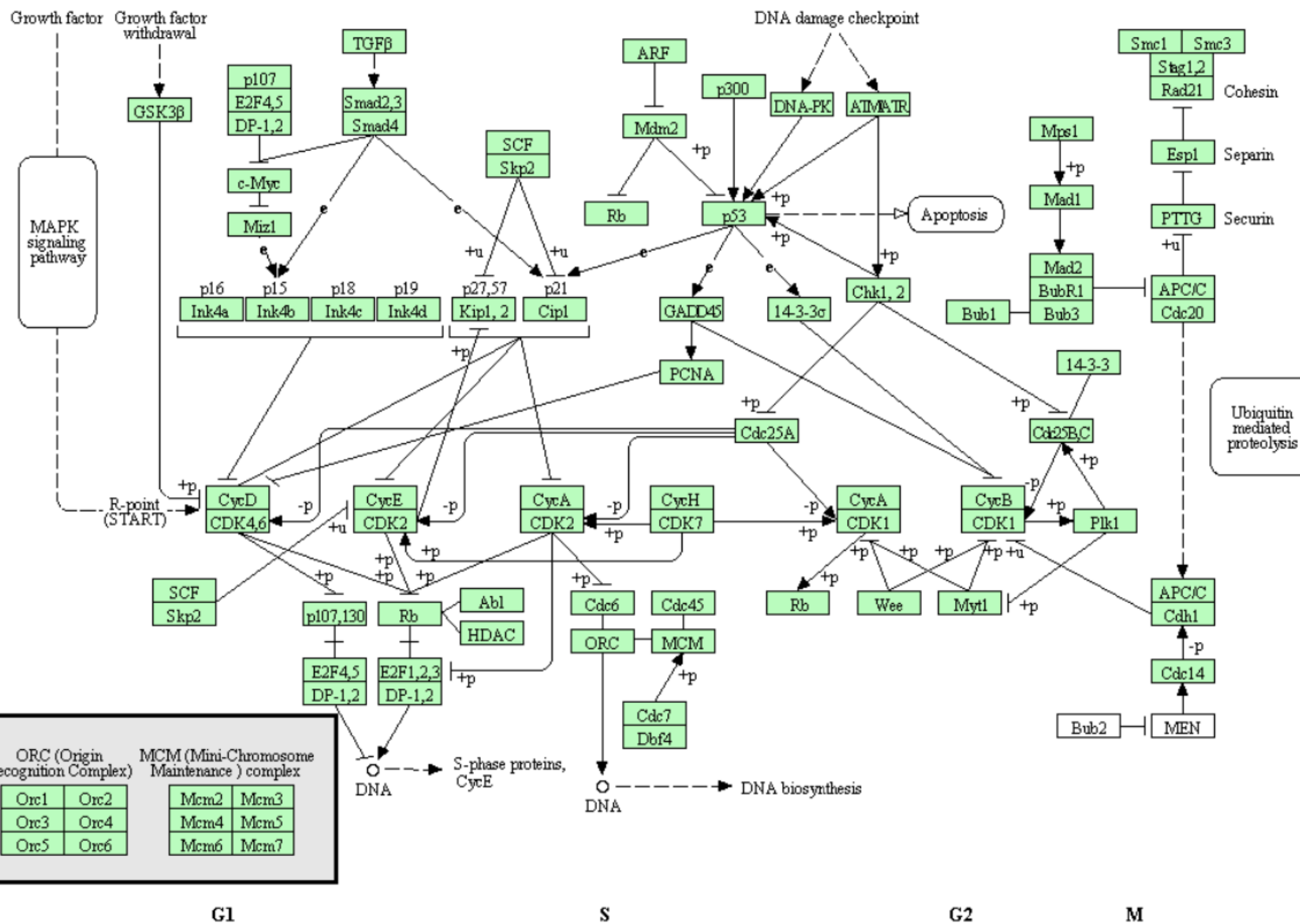
Main types of biological pathways:

- Metabolic pathways

- Gene regulation pathways

- Signal transduction pathways

CELL CYCLE



Pathway databases

KEGG PATHWAY

BioCarta

BioCyc

Protein ANalysis THrough Evolutionary Relationships (PANTHER)

Pathway Interaction Database (PID)

Reactome

<http://www.pathguide.org/>

Pathguide the pathway resource list

Home BioPAX CBio MSKCC

Navigation

- Protein-Protein Interactions
- Metabolic Pathways
- Signaling Pathways
- Pathway Diagrams
- Transcription Factors / Gene Regulatory Networks
- Protein-Compound Interactions
- Genetic Interaction Networks
- Protein Sequence Focused
- Other

Search

Organisms:

Availability:

Standards:

Analysis

- Statistics
- Database Interactions

Contact

Comments, Questions, Suggestions are Always Welcome!

Complete Listing of All Pathguide Resources

Pathguide contains information about **702** biological pathway related resources and molecular interaction related resources. Click on a link to go to the resource home page or 'Details' for a description page. Databases that are free and those supporting BioPAX, CellML, PSI-MI or SBML standards are respectively indicated.

If you know of a pathway resource that is not listed here, or have other questions or comments, please [send us an e-mail](#).

News

Major new update of Pathguide
September 2017
We now have information for over 690 resources!

Major new update of Pathguide
August 2013
We now have information about ~550 resources!

Protein-Protein Interactions

Database Name (Order: alphabetically by web popularity)	Full Record	Availability	Standards
2P2ldb - The Protein-Protein Interaction Inhibition Database	Details	Free	
3D-Interologs - 3D-Interologs	Details	Free	
3DID - 3D interacting domains	Details	Free	
ACSN - Atlas of Cancer Signalling Network	Details	Free	BioPAX
ADAN - Prediction of protein-protein interaction of modular domains	Details	X	
AHD2.0 - Arabidopsis Hormone Database 2.0	Details	Free	
AllFuse - Functional Associations of Proteins in Complete Genomes	Details	X	
aMAZE - Protein Function and Biochemical Pathways Project	Details	X	
ANAP - Arabidopsis Network Analysis Pipeline	Details	Free	
ANIA - ANnotation and Integrated Analysis of the 14-3-3 interactome	Details	Free	
AnimalTFDB - Animal Transcription Factor Database	Details	Free	
AntiJen - AntiJen a Kinetic, Thermodynamic and Cellular Database	Details	Free	
APID - Agile Protein Interactomes DataServer	Details	Free	PSI-MI
Araport - Arabidopsis Information Portal	Details	Free	
ARN - The Autophagy Regulatory Network	Details	X	PSI-MI BioPAX

KEGG PATHWAY

<http://www.kegg.jp/kegg/>



KEGG: Kyoto Encyclopedia of Genes and Genomes

Data-oriented entry points

KEGG PATHWAY	KEGG pathway maps [Pathway list]
KEGG BRITE	BRITE functional hierarchies [Brite list]
KEGG MODULE	KEGG modules [Module list]
KEGG DISEASE	Human diseases [Cancer Infectious disease]
KEGG DRUG	Drugs [ATC drug classification]
KEGG ORTHOLOGY	Ortholog groups [KO system]
KEGG GENOME	Genomes [KEGG organisms]
KEGG GENES	Genes and proteins Release history
KEGG COMPOUND	Small molecules [Compound classification]
KEGG REACTION	Biochemical reactions [Reaction modules]

Kanehisa *et al.*, *Nucleic Acids Res.*, 2012

Kanehisa and Goto, *Nucleic Acids Res.*, 2000

KEGG PATHWAY



KEGG Pathway Maps

[[Brite menu](#) | [Download htext](#)]

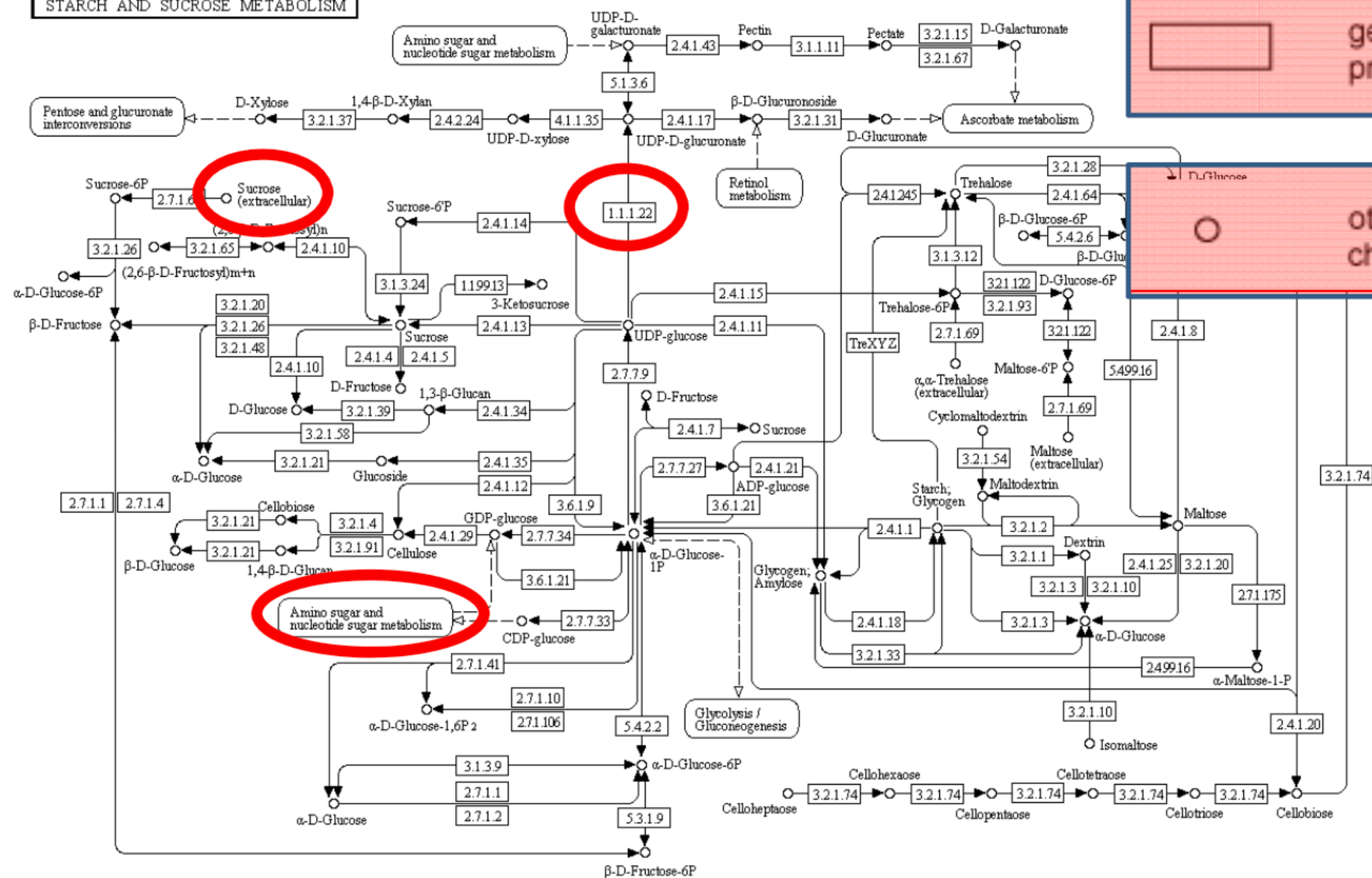
KEGG pathway maps



☐ One-click mode

- ▶ **Metabolism**
- ▶ **Genetic Information Processing**
- ▶ **Environmental Information Processing**
- ▶ **Cellular Processes**
- ▶ **Organismal Systems**
- ▶ **Human Diseases**
- ▶ **Drug Development**

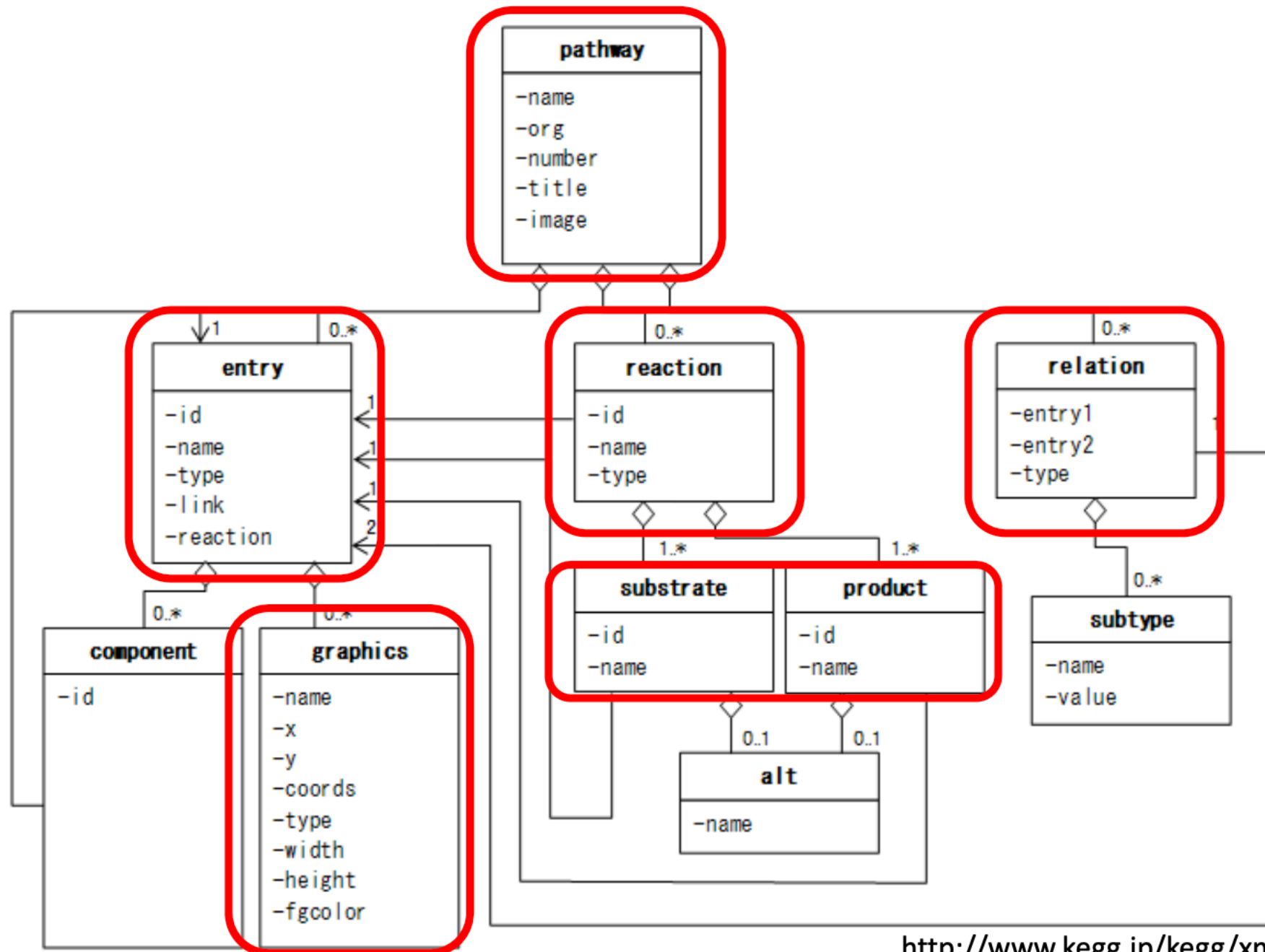
STARCH AND SUCROSE METABOLISM



gene product, mostly protein but including RNA

other molecule, mostly
chemical compound

KEGG Markup Language (KGML) format



http://www.kegg.in/kegg/xml/docs/kgml_class.png

KGML format

```
<pathway name="path:hsa04110" org="hsa" number="04110"
  title="Cell cycle"
  image="http://www.kegg.jp/kegg/pathway/hsa/hsa04110.png"
  link="http://www.kegg.jp/kegg-bin/show_pathway?hsa04110">
  <entry id="1" name="hsa:1029" type="gene"
    link="http://www.kegg.jp/dbget-bin/www_bget?hsa:1029">
    <graphics name="CDKN2A, ARF, CDK4I, CDKN2, CMM2, INK4, INK4A, MLM, MTS-1, MTS1, P14, P14ARF, P16, P16-
NK4, P16INK4A, P19, P19ARF, TP16" fgcolor="#000000" bgcolor="#BFFFBF"
      type="rectangle" x="532" y="124" width="46" height="17"/>
    </entry>
    <entry id="2" name="hsa:51343" type="gene"
      link="http://www.kegg.jp/dbget-bin/www_bget?hsa:51343">
      <graphics name="FZR1, CDC20C, CDH1, FZR, FZR2, HCDH, HCDH1" fgcolor="#000000" bgcolor="#BFFFBF"
        type="rectangle" x="919" y="536" width="46" height="17"/>
      </entry>
      <entry id="3" name="hsa:4171 hsa:4172 hsa:4173 hsa:4174 hsa:4175 hsa:4176" type="gene"
        link="http://www.kegg.jp/dbget-bin/www_bget?hsa:4171+hsa:4172+hsa:4173+hsa:4174+hsa:4175+hsa:4176">
        <graphics name="MCM2, BM28, CCNL1, CDCL1, D3S3194, MITOTIN, cdc19..." fgcolor="#000000" bgcolor="#BFFF
          type="rectangle" x="553" y="556" width="46" height="17"/>
        </entry>
        <entry id="4" name="hsa:23594 hsa:23595 hsa:4998 hsa:4999 hsa:5000 hsa:5001" type="gene"
          link="http://www.kegg.jp/dbget-bin/www_bget?hsa:23594+hsa:23595+hsa:4998+hsa:4999+hsa:5000+hsa:5001">
          <graphics name="ORC6, ORC6L..." fgcolor="#000000" bgcolor="#BFFFBF"
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<reaction id="85" name="rn:R00026" type="irreversible">
  <substrate id="222" name="cpd:C00185"/>
  <product id="193" name="cpd:C00221"/>
</reaction>
<reaction id="86" name="rn:R03920" type="reversible">
  <substrate id="176" name="cpd:C02336"/>
  <product id="194" name="cpd:C05345"/>
</reaction>
<reaction id="93" name="rn:R00802" type="irreversible">
  <substrate id="186" name="cpd:C00089"/>
  <product id="191" name="cpd:C00267"/>
  <product id="176" name="cpd:C02336"/>
</reaction>
<reaction id="95" name="rn:R00802" type="irreversible">
  <substrate id="186" name="cpd:C00089"/>
  <product id="191" name="cpd:C00267"/>
  <product id="176" name="cpd:C02336"/>
</reaction>
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<relation entry1="49" entry2="47" type="GRel">
  <subtype name="expression" value="--&gt;"/>
</relation>
<relation entry1="49" entry2="32" type="GRel">
  <subtype name="expression" value="--&gt;"/>
</relation>
<relation entry1="47" entry2="23" type="PPrel">
  <subtype name="activation" value="--&gt;"/>
</relation>
<relation entry1="36" entry2="48" type="PPrel">
  <subtype name="inhibition" value="--|"/>
</relation>
<relation entry1="36" entry2="49" type="PPrel">
  <subtype name="inhibition" value="--|"/>
</relation>
<relation entry1="37" entry2="49" type="PPrel">
  <subtype name="activation" value="--&gt;"/>
</relation>
```




THANK YOU